FoodGO — Prioritised product backlog and non-functional requirements

Requirements artifact for the CS 488 midterm. Source of truth: the Lab 3 specification (`docs/lab3-specification`). Every story carries the requirement IDs it implements, so it traces to the spec, the SDD (`docs/lab4-sdd`) and the code (`docs/requirements/TRACEABILITY.md`).

- **Priority:** MoSCoW. **Estimate:** story points (Fibonacci), estimated by the team with AI-suggested first values then corrected.
- **Status:** ✅ Done in the midterm MVP · ➡️ Planned for a later sprint · ⛔ Blocked on a stakeholder decision (spec §19.7).

Epics

Epic	Stories	FRs
E0 Service area	US-32	FR-32, BR-17 (team decision, ADR-0007)
E1 Accounts & access	US-01 – US-05	FR-01–FR-04, FR-29, FR-30, BR-05, BR-16
E2 Restaurant & menu	US-06 – US-09	FR-05–FR-08, FR-12–FR-13 settings, BR-03
E3 Ordering	US-10 – US-13	FR-09–FR-15, BR-01, BR-02, BR-12
E4 Verification	US-14 – US-15	FR-16, FR-17, BR-04, BR-15
E5 Delivery	US-16 – US-20	FR-18–FR-24, BR-06–BR-11
E6 Payment	US-21 – US-23	FR-25–FR-28, BR-12, BR-13
E7 AI-assisted planning	US-24	FR-31 (PROPOSED), BG-03, BG-04
E8 Platform quality	US-25 – US-28	NFRs below

User stories

Sorted by priority: Must first, then Should, then Could; within a priority, by dependency order (story ID).

Rank	ID	User story	Acceptance criteria (spec §17)	Req.	MoSCoW	SP	Status
1	US-01	As **restaurant staff** I want to register a customer in person so that only people I verified can order from me.	Name, phone, organization captured; staff must tick "identity checked"; duplicate registration refused; customer receives an SMS.	FR-01–FR-04, BR-05, BR-16	Must	5	✅

Rank	ID	User story	Acceptance criteria (spec §17)	Req.	MoSCoW	SP	Status
2	US-02	As a **customer** I want to sign in with my phone number and a one-time code so that I don't need a password.	Code sent by SMS (sandbox); single use; expires after 5 min; 5 wrong attempts lock the code; unregistered numbers refused.	FR-01, FR-03, BR-05	Must	5	✅
3	US-03	As **restaurant staff / delivery person / admin** I want to sign in with a password.	bcrypt hashes only; same answer for unknown identity and wrong password; session expires after 12 h.	FR-29, FR-30, §12	Must	3	✅
4	US-04	As the **platform administrator** I want to create restaurants and provision their staff accounts.	Staff account bound to exactly one restaurant; duplicate identity refused; admins cannot be created from the app.	FR-29	Must	3	✅
5	US-05	As **restaurant staff** I want to create accounts for my own delivery team.	Account bound to my restaurant; never visible to other restaurants.	FR-30, FR-18, BR-06	Must	2	✅
6	US-06	As a **customer** I want to see only the restaurants that registered me.	Unregistered or ineligible restaurants are not listed and not reachable.	FR-05, BR-04, BR-16	Must	2	✅
7	US-07	As a **customer** I want to browse a menu and see which dishes are available today.	Unavailable dishes are shown but cannot be added.	FR-06, FR-08, BR-02	Must	2	✅
8	US-08	As **restaurant staff** I want to add, edit, remove dishes and toggle availability.	Price change never alters placed orders (D-02); removal is soft.	FR-07, FR-08, BR-03	Must	3	✅
9	US-09	As **restaurant staff** I want to set my daily ordering deadline and payment options.	HH:MM validated; at least one payment method stays on; changing the deadline never changes placed orders (D-04).	FR-12, FR-13, FR-26	Must	2	✅

Rank	ID	User story	Acceptance criteria (spec §17)	Req.	MoSCoW	SP	Status
10	US-10	As a **customer** I want to choose dishes and quantities and see the total before ordering.	Total = Σ quantity × price, exact in millimes; 1–50 per dish.	FR-09–FR-11	Must	3	✅
11	US-11	As a **customer** I want to place my order before the deadline and get an order number.	Accepted before the deadline, refused at/after it with ORDER-001; order number FG-XXXXXX; retry with same Idempotency-Key never duplicates.	FR-12–FR-14, BR-01	Must	5	✅
12	US-12	As **restaurant staff** I want new orders to appear in my queue immediately.	Live board; only my restaurant's orders; customer verification info shown.	FR-15, FR-16, BR-09	Must	3	✅
13	US-14	As **restaurant staff** I want the information I need to verify a customer.	Only for customers registered at my restaurant.	FR-16, BR-15	Must	2	✅
14	US-15	As **restaurant staff** I want to mark a customer eligible or not for my organization rule.	Ineligible customers lose access to my restaurant only.	FR-17, BR-04	Must	2	✅
15	US-16	As **restaurant staff** I want to assign an order to one of my delivery people.	Only my own couriers; assignment exists before dispatch; no double assignment.	FR-19, BR-06, BR-10	Must	3	✅
16	US-17	As a **delivery person** I want to see only the orders assigned to me with the details I need.	Scoped query; destination, phone, items, cash to collect.	FR-20, FR-21, BR-08, BR-15	Must	3	✅
17	US-18	As a **delivery person** I want to start a delivery and confirm it when delivered.	Confirm only after start, only by the assignee, only once.	FR-22, FR-23, BR-11	Must	3	✅

Rank	ID	User story	Acceptance criteria (spec §17)	Req.	MoSCoW	SP	Status
18	US-21	As a **customer** I want to choose cash on delivery or e-payment.	Only methods the restaurant enabled.	FR-25, FR-26	Must	2	✅
19	US-23	As the **restaurant** I want every order's payment status recorded.	Payment row created with the order; cash marked paid when the courier confirms collection; order shows a projection of it.	FR-28, BR-12	Must	3	✅
20	US-26	As a **developer** I want CI to lint and test every pull request.	GitHub Actions on Node 22 and 24; coverage ≥ 60 %.	Course rubric	Must	2	✅
21	US-27	As a **developer** I want API documentation.	OpenAPI 3 file + Swagger UI at `/api/docs`.	Course rubric	Must	3	✅
22	US-28	As the **team** we want before/after metrics for refactoring and performance.	Report with complexity, lint, coverage, latency.	Course rubric	Must	5	✅
23	US-32	As **FoodGO** we want to serve only Lac 3 / Zone Industrielle Le Kram so that employees there stop driving out at lunchtime.	Admin must place every restaurant and staff every workplace in the area (zone `LAC3_ZI_KRAM`); any other value refused (VAL-001, DB CHECK); an order is refused unless restaurant and workplace are both in the area (ORDER-003); zone shown to customers, staff and couriers.	FR-32, BR-17	Must	3	✅
24	US-13	As **restaurant staff** I want to move an order into preparation.	Only legal transitions (State pattern).	BR-14	Should	1	✅
25	US-19	As a **customer** I want to follow my order's progress live.	Status updates without refreshing; I never see other people's orders.	FR-24, BR-14, BR-15	Should	3	✅

Rank	ID	User story	Acceptance criteria (spec §17)	Req.	MoSCoW	SP	Status
26	US-22	As a **customer** I want to pay electronically through a supported Tunisian provider.	Sandbox adapter now; declined payment rolls back the order (PAY-001); provider down → PAY-002. Real provider blocked on Q-22.	FR-27, BR-13	Should	5	✅ sandbox / ⛔ real provider
27	US-24	As **restaurant staff** I want a forecast of tomorrow's dishes so that I prepare the right quantities.	Per-dish expected quantity, range, confidence; accuracy shown; advice only, never changes orders; optional Claude summary.	FR-31 (PROPOSED), BG-03, BG-04	Should	8	✅
28	US-25	As the **administrator** I want to monitor today's orders and security events.	Read-only monitor with audit trail.	BG-09, §15	Should	3	✅
29	US-31	As the **team** we want the app deployed on a public host.	HTTPS, persistent volume for SQLite.	§7 availability	Should	3	➡️ SOON
30	US-33	As a **developer** I want Order Management to expose intention-revealing transitions (`markAssigned`, `dispatch`, `markDelivered`) that load the assignment themselves.	The State guards (BR-10, BR-11) no longer trust a context built by the caller; `applyTransition` becomes private. Raised by the Lab 5 peer review.	BR-10, BR-11, BR-14	Should	3	➡️ SOON
31	US-20	As a **delivery person** I want to record failed deliveries.	—	Q-07, Q-20	Could	3	⛔
32	US-29	As a **customer** I want to cancel an order before preparation.	—	Q-04–Q-06	Could	5	⛔
33	US-30	As the **platform** we want real SMS and e-mail notifications.	Adapter swap only.	§5.2.2, Q-22	Could	5	➡️ SOON

Sprint 1 velocity (midterm MVP): 89 story points done out of 108 in the backlog (including US-32, added on 1 Nov 2026 when the team narrowed the service area, ADR-0007).

Non-functional requirements

The specification states quality attributes without numbers (§7). The targets below are the spec's own **Proposed** targets; the "How we check it" column says what is actually verified today.

ID	Attribute	Requirement / target	How we check it	Status
NFR-01	Performance	95 % of API requests < 500 ms, 99 % < 1.5 s; order submission < 2 s	`scripts/benchmark.js` (see refactor report)	✅ measured
NFR-02	Security	Authenticated roles, restaurant-scoped authorisation, hashed credentials, rate-limited sign-in, safe errors, security headers	Integration tests (AUTH-001/002 cases), helmet, express-rate-limit	✅
NFR-03	Privacy	Customer data visible only to authorised parties (BR-15); no personal data sent to the LLM	Scope tests, prompt privacy test	✅
NFR-04	Reliability	No lost, duplicate or half-written orders	Transactions T1–T3, Idempotency-Key, rollback tests	✅
NFR-05	Maintainability	Business rules separated from UI and persistence; lint clean; complexity ≤ 10 per function	Layered architecture, ESLint `complexity` rule, CI	✅
NFR-06	Testability	Each feature has unit + integration tests; ≥ 60 % line coverage	`npm run coverage` in CI: 97.3 % of all server code (`src/` , browser files excluded); 36 unit + 83 integration tests	✅
NFR-07	Usability	Simple ordering in ≤ 4 screens; deadline visible before ordering	Customer UI	✅
NFR-08	Accessibility	Labelled controls, keyboard focus, reduced motion; WCAG 2.2 AA target	Manual review	➡️ full audit
NFR-09	Observability	Audit log of security-relevant actions; health endpoint	`audit_event` , `/health` , admin monitor	✅
NFR-10	Interoperability	Payment, SMS/OTP, notification, LLM behind adapters	Adapter unit tests	✅

ID	Attribute	Requirement / target	How we check it	Status
NFR-11	Availability	99.5 % during service hours	Not measurable before deployment	➡️
NFR-12	Portability	Runs with Node.js 22+ only, no external service required	Fresh-clone check in README	✅

AI-assisted elicitation record

As the course requires, this is the prompt used to turn the specification into stories, followed by what the team changed.

Prompt (Claude, 1 Nov 2026):

> Act as a product analyst. Given: the FoodGO Software Specification (30 functional requirements FR-01–FR-30, 16 business rules, acceptance criteria in §17, and the unresolved items in §19). Produce: (1) user stories with acceptance criteria for every FR, grouped into epics; (2) the non-functional requirements with measurable targets where the specification proposes them; (3) a MoSCoW priority and a story-point estimate per story. Keep every requirement ID. Do not invent behaviour for unresolved items: mark them blocked.

What the team changed after review:

- Merged FR-01–FR-04 into one registration story (US-01): they are one in-person action at the counter.
- Added the BR-05 "identity checked in person" checkbox to US-01's criteria; the AI draft only collected data.
- Raised US-22 (e-payment) from Could to Should, but split its status: the sandbox is done, the real provider stays blocked on Q-22.
- Added US-24 (AI forecast) after the team withdrew assumption A-13; the AI draft had correctly left AI out because the spec excluded it.
- Re-estimated three stories the AI under-estimated (US-02 OTP 3→5, US-11 ordering 3→5, US-24 forecast 5→8).